

The Hyper Drumhead: A Musical Instrument For The Audio/Visual Manipulation Of Sound Waves

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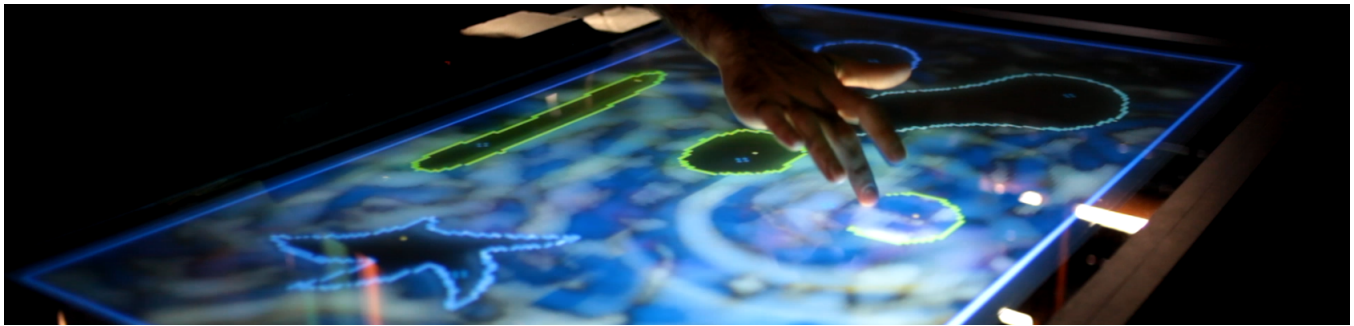


Figure 1: Sound waves propagating from an excitation point on the 5 ms latency, multi-touch surface of the Hyper Drumhead.

ABSTRACT

The Hyper Drumhead is a novel digital musical instrument that allows for the visualization and the manipulation of sound waves. At its core, a GPU-accelerated physical model computes in real-time the propagation of sound waves in two dimensions, allowing for the audio/visual simulation of massive domains. Every time the musician touches the surface of the instrument, an excitation signal is injected into the simulation domain, triggering wave propagation in all directions. By drawing boundaries in the domain and modifying the acoustic parameters of the simulated medium, the musician may generate and modulate reflections and resonances, effectively shaping the timbre of the resulting sound. In this short paper, we describe the main components of the instrument, including the control interface and the underlying numerical simulation.

CCS CONCEPTS

• **Computing methodologies** → **Massively parallel and high-performance simulations**; • **Applied computing** → **Sound and music computing**; • **Human-centered computing** → **Graphics input devices**.

KEYWORDS

audio/visual, musical instrument, acoustic simulation, GPU

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SIGGRAPH '22 Real-Time Live!, August 07-11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9368-3/22/08.

<https://doi.org/10.1145/3532833.3538686>

ACM Reference Format:

Victor Zappi, Mike Frengel, Andrew Allen, and Sidney Fels. 2022. The Hyper Drumhead: A Musical Instrument For The Audio/Visual Manipulation Of Sound Waves. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Real-Time Live! (SIGGRAPH '22 Real-Time Live!)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532833.3538686>

1 INTRODUCTION

When the surface of an object is hit, a transverse wave starts propagating from the the impact location, triggering a vibration that may extend to the full body of the object. This mechanism is at the core of most percussion musical instruments. The vibration displaces the air particles adjacent to the instrument's surface, in turn generating an acoustic wave that we can perceive as sound/music. The sonic nuances of percussion instruments are then characterized by a variety of more complex non-linear phenomena, that affect the traveling wave and depend on the design details of the specific build. Yet, by only modeling wave propagation on the instrument's surface, it is possible to approximate with a good degree of precision the signature sounds of several of these instruments, including the acoustic outcomes of different shapes, sizes, materials and types of excitation [Sosnick and Hsu 2010].

In this work we present the Hyper Drumhead, a novel digital musical instrument that simulates in real-time 2D wave propagation, and provides control over the geometry and the acoustic properties of the modeled medium. The main body of the instrument consists of a tabletop interface, composed of a 42 inch retro-projected multi-touch glass display hosted in a custom raised frame. Every time the musician touches the glass surface of the instrument, an excitation signal is injected into the underlying acoustic simulation;

this results into the generation of propagating transverse waves that depart from the touch location (1). Such waves are rendered in two modalities: visually, through a color scale that goes from white to black to blue (where white shows positive displacement, blue negative displacement and black rest position); and sonically, via a set of movable virtual microphones that turn the local displacement into an audio signal.

2 NUMERICAL SIMULATION

At its core, the Hyper Drumhead solves numerically the 2D acoustic wave equation. The solver leverages a finite-different time-domain (FDTD) scheme to calculate the local surface displacement in every grid point of the simulated domain. More details about the scheme and the underlying equations can be found in [Zappi et al. 2017].

When running on a modern CPU, the proposed FDTD scheme can solve in real-time domains spanning no more than 20×20 grid points [Renney et al. 2019; Zappi et al. 2017]. This is enough to synthesize sounds that closely resemble instruments like membranes, cymbals and plates, yet much larger domains are required for expressive musical interaction [Sosnick and Hsu 2010; Zappi et al. 2017]. To achieve the simulation of large domains, the Hyper Drumhead's solver is designed to run in parallel over the streaming multiprocessors of GPUs. In particular, the solver is implemented as a set of OpenGL shaders, serving both as video and audio renderers. This technology draws inspiration from the work of Allen and Raghuvanshi [Allen and Raghuvanshi 2015], which consisted of a system for the real-time modeling of 2D aerophones. We appropriated this approach for the case of percussion instruments and improved upon it, to target expressive real-time musical interactions.

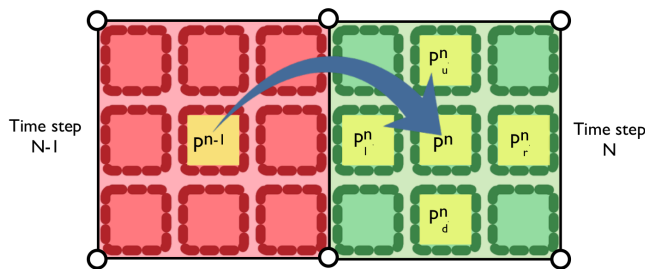


Figure 2: Fragment access pattern.

The local displacement is computed and stored on a texture, where each fragment represents a single FDTD grid point. To compute the next displacement on a fragment (p^{n+1}), the solver needs access to the current and the previous value of the same point on the grid (p^n and p^{n-1}), as well as the current values of the top, bottom, left and right neighbor points ($p^n_u, p^n_d, p^n_l, p^n_r$). For this reason, the texture is large enough to store side by side two copies of the domain, containing displacement values for time step n and time step $n - 1$, respectively. A first fragment shader uses a frame buffer object to access both sides of the texture (Figure 2) and to overwrite time step $n - 1$ with the displacement values for next time step ($n + 1$). This operation happens at audio rate (44100 Hz) and in parallel across the full domain, with the number of concurrent threads

depending on the specifications of the employed GPU. Then, the updated side becomes n and, vice versa, the other becomes $n - 1$. A second fragment shader running at a much lower rate reads the most updated side of the texture, maps each displacement values to the color scale and fills the render buffer with the result.

The whole set of shaders is highly optimized and allows for expressive musical interaction on massive domains. When running on an Nvidia GeForce GTX 1080, the Hyper Drumhead can span a domain of 238×133 grid points, split in up to 8 *areas*, each characterized by distinct excitation type and virtual microphone, as well as shape and type of material.

3 MUSICAL INTERACTION

The multi-touch interface of the Hyper Drumhead captures a variety of gestures that can be leveraged for making music. Some of these gestures can be triggered via a MIDI footswitch placed under the screen. The simplest interaction paradigm consists of the release of an impulse in every location of the surface that is touched by the musician. By acting on the footswitch, the musician can then use their fingers to draw and erase boundary points directly on the domain; these allow for the creation of arbitrary geometrical layouts, that generate reflections and resonances, and shape the timbre of the instrument. Furthermore, with a pinching gesture the simulation parameters of the touched area can be modified in real-time, to change the acoustics of the modeled material. Altogether, these mechanisms permit to precisely match the sound of a target percussion instrument.

The surface of the Hyper Drumhead senses pressure too. By default, this value is used to determine the intensity of each excitation but, by acting on the footswitch, it can be employed to modulate the acoustic properties of the touched area. This allows for the execution of more advanced playing techniques.

The instrument supports two further types of excitation that can replace the basic impulse. The first one, allows the musician to excite the touched area with any pre-recorded waveform (loaded through standard audio files), while the second one injects into the domain live audio streams captured via line inputs or microphones. These types of *continuous excitation* unlock novel interaction paradigms, as the musician can move around the domain steady sound streams and explore in real-time how their spectrum is filtered by different shapes and materials.

4 CONCLUSION

The Hyper Drumhead leverages massive GPU acceleration to run highly interactive numerical acoustic simulations. The result is an innovative audio/visual instrument, that can be used both as a synthesizer and as a processor to make a wide variety of music.

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